

# ENCN205

Applied Data Analysis for Civil and Natural Systems

## LECTURE 1

Why Location Matters — Introduction to GIS

University of Canterbury | Te Whare Wānanga o Waitaha

**Dr Alberto Ardid**

Lecturer / Assistant Professor  
Department of Civil and Environmental Engineering  
University of Canterbury, New Zealand

E319 |  
[alberto.ardid@canterbury.ac.nz](mailto:alberto.ardid@canterbury.ac.nz)

# "Where is it?" is one of the most powerful question in engineering.

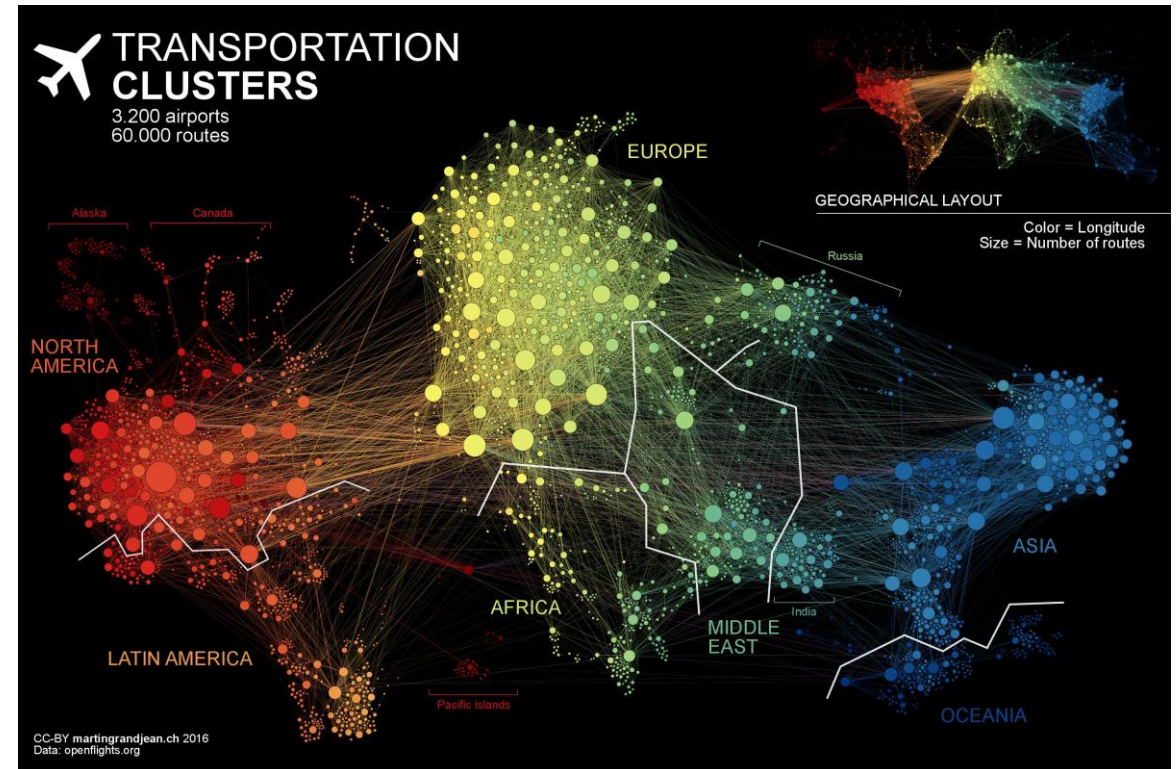
Where should we build the new water intake?

Which buildings will flood if the river rises 2 metres?

Where are the weakest links in the pipe network?

Which communities are most vulnerable to earthquake damage?

**3,200 airports and 60,000 routes - spatial data reveals patterns invisible in spreadsheets.**



# What is spatial thinking?

**Spatial thinking** is the process of **problem solving** via the **coordinated use of space, representation, and reasoning**. Spatial thinking uses representations to help us remember, understand, reason, and communicate about **the properties of and relations between objects represented in space**, whether or not those objects themselves are inherently spatial.

National Academies of Sciences, Engineering, and Medicine. 2006. Learning to Think Spatially. Washington, DC: The National Academies Press. <https://doi.org/10.17226/11019>.





# The First GIS Analysis

London, 1854 — a cholera outbreak was killing hundreds; , and the prevailing theory says the disease spreads through bad air.

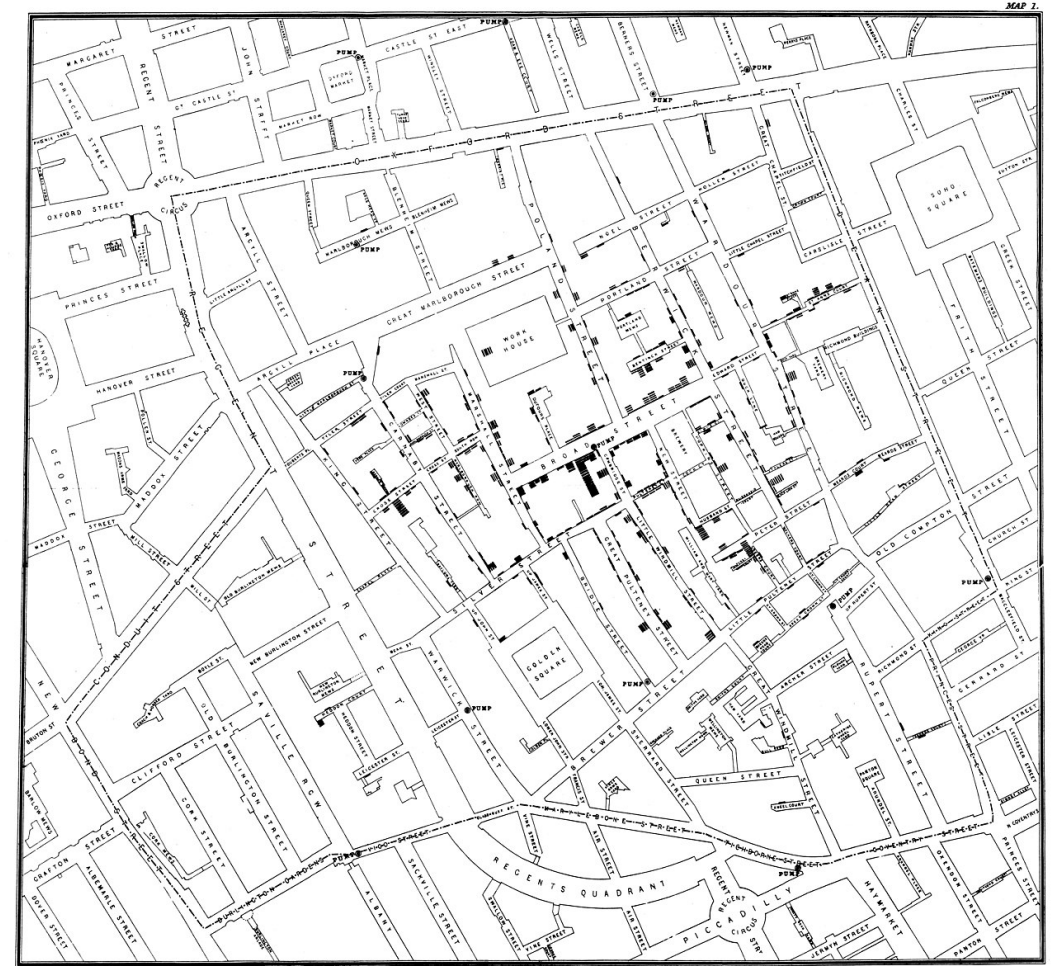
Dr John Snow plotted each death as a dot on a street map. The dots clustered around one location: the Broad Street pump.

Snow convinced authorities to remove the pump handle. The epidemic subsided — spatial analysis saved lives.

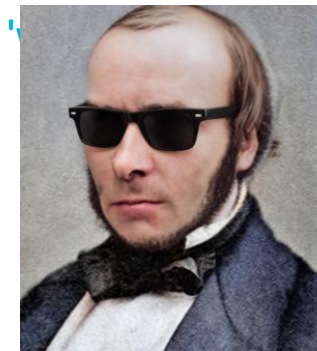
**A dot map solved a public health crisis 170 years ago — the same logic powers modern GIS.**

Snow's map didn't show the contaminated sewage source; he still needed the mechanism to prove his case.  
**The map pointed; the mechanism convinced.**

A map is evidence, not proof.



**Saved fictional  
people from  
fictional zombies**



**Saved real people  
from a real  
cholera outbreak**

Always ask: 'What is the mechanism showing?'

# Location Carries History: The Alabama Black Belt

**A coastline from 100 million years ago is still visible in modern election maps.**

Cretaceous seas deposited chalk in a crescent across Alabama

Chalk weathered into fertile, dark 'Blackland Prairie' soil

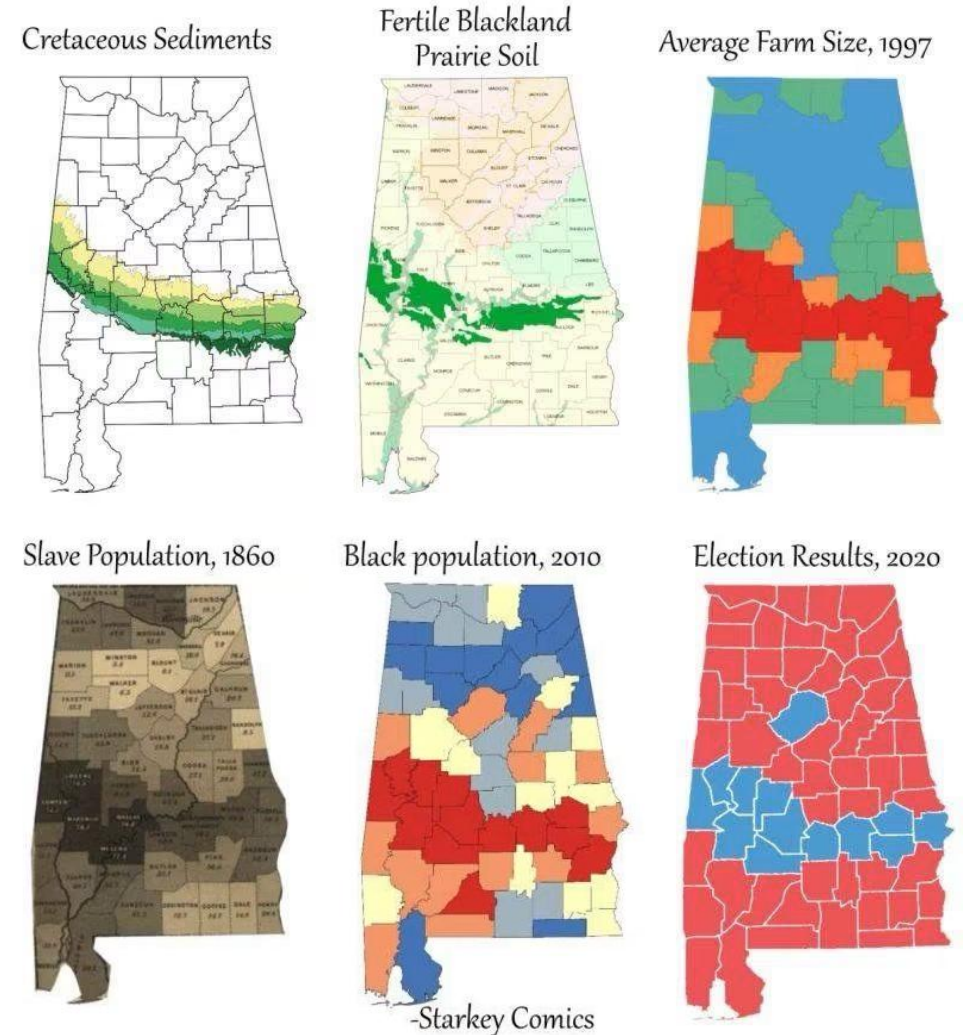
1800s: fertile soil → cotton plantations → enslaved population concentrated in the crescent

Today: Black communities remain concentrated there — and the 2020 election map traces the ancient shoreline

**Geology shapes soils, soils shape land use, land use shapes society - spatial patterns persist for centuries.**

**The point for you as engineers: spatial patterns persist, sometimes for centuries, and layers of spatial data let us connect causes and effects across time.**

How a coastline 100 million years ago influences modern election results in Alabama



Source: Starkey Comics

# What is GIS?

A Geographic Information System (GIS) is a framework for:

Collecting spatial and attribute data

Storing and managing geographic datasets

Analysing spatial relationships and patterns

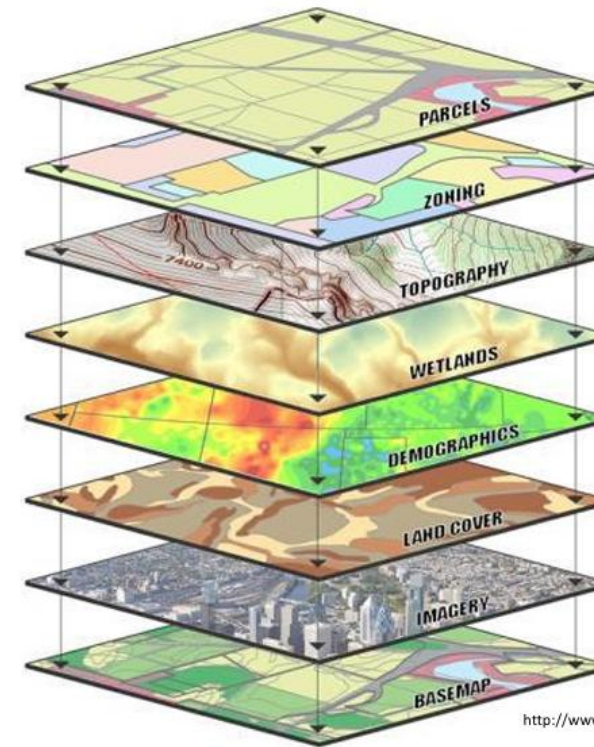
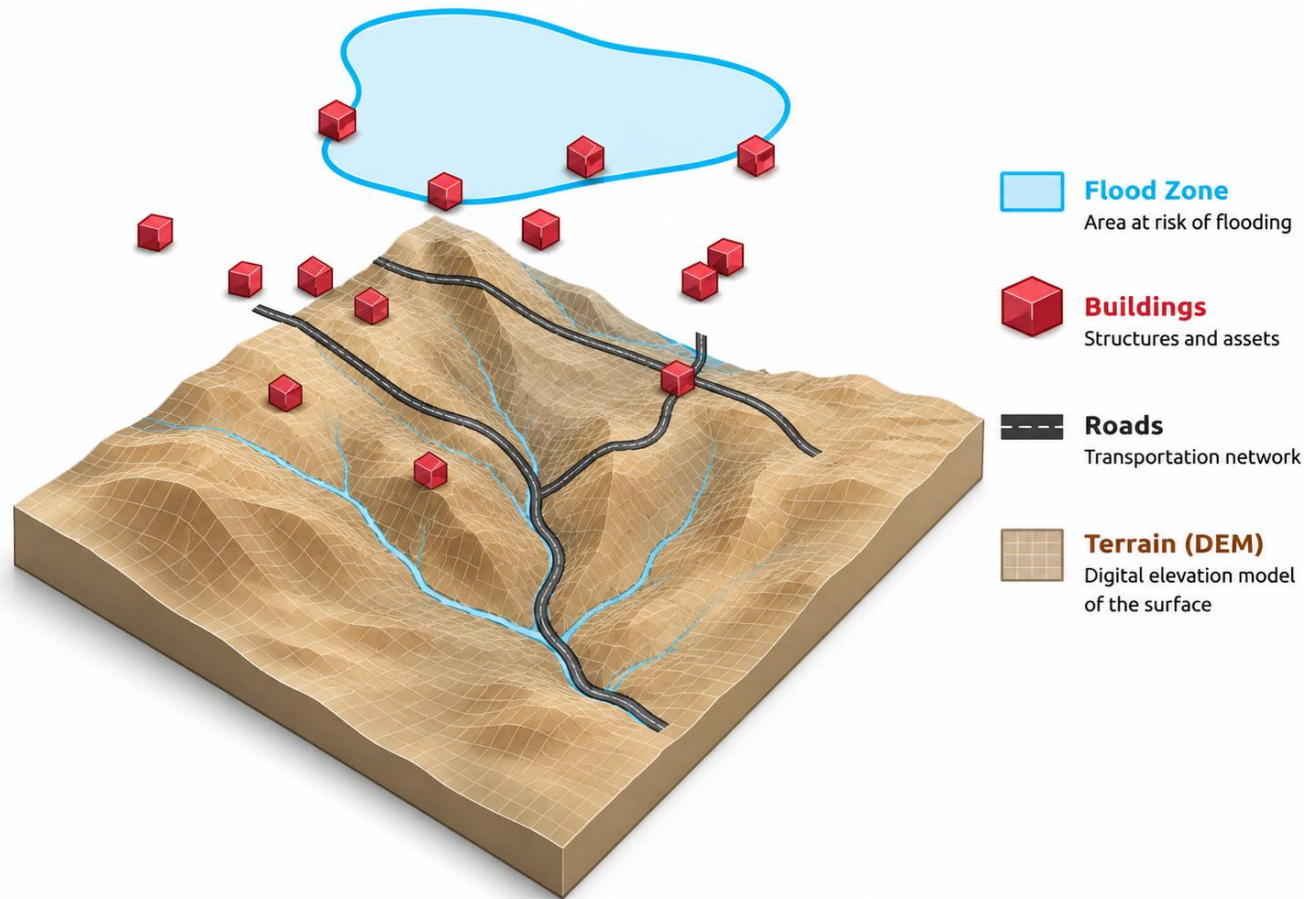
Visualising results as maps and reports

**It is both a tool (software) and a science (spatial thinking: knowing which question to ask and which analysis answers it).**

**Software versions come and go; the thinking is what makes you valuable as an engineer.  
(that's what I actually want you to leave this course with)**



# GIS: Layers of Spatial Information



## GIS DATA LAYERS

Many different types of data can be integrated into a GIS and represented as a map layer.

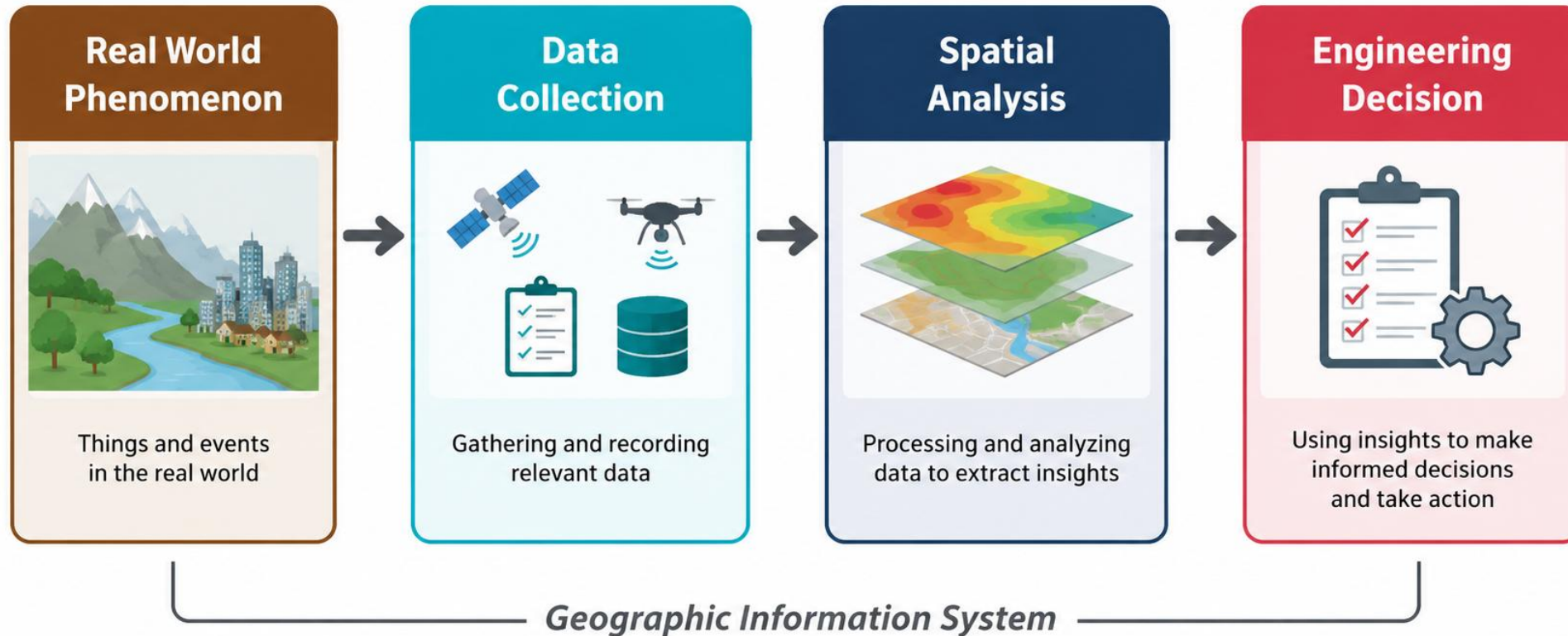
Examples can include: streets, parcels, zoning, flood zones, client locations, competition, shopping centers, office parks, demographics, etc.

When these layers are drawn on top of one another, undetected spatial trends and relationships often emerge. This allows us to gain insight about relevant characteristics of a location.

<http://www.co.ontario.ny.us/images/pages/N1176/gisdata.jpg>

Each layer represents a different aspect of the real world — GIS combines them to answer engineering questions.  
**Layers plus spatial relationships equals insight.**

# The GIS Workflow *(how does GIS actually fit into engineering work? )*



**Key principle: GIS connects real-world observations to engineering decisions.**

The quality of the decision depends on the quality of every step.

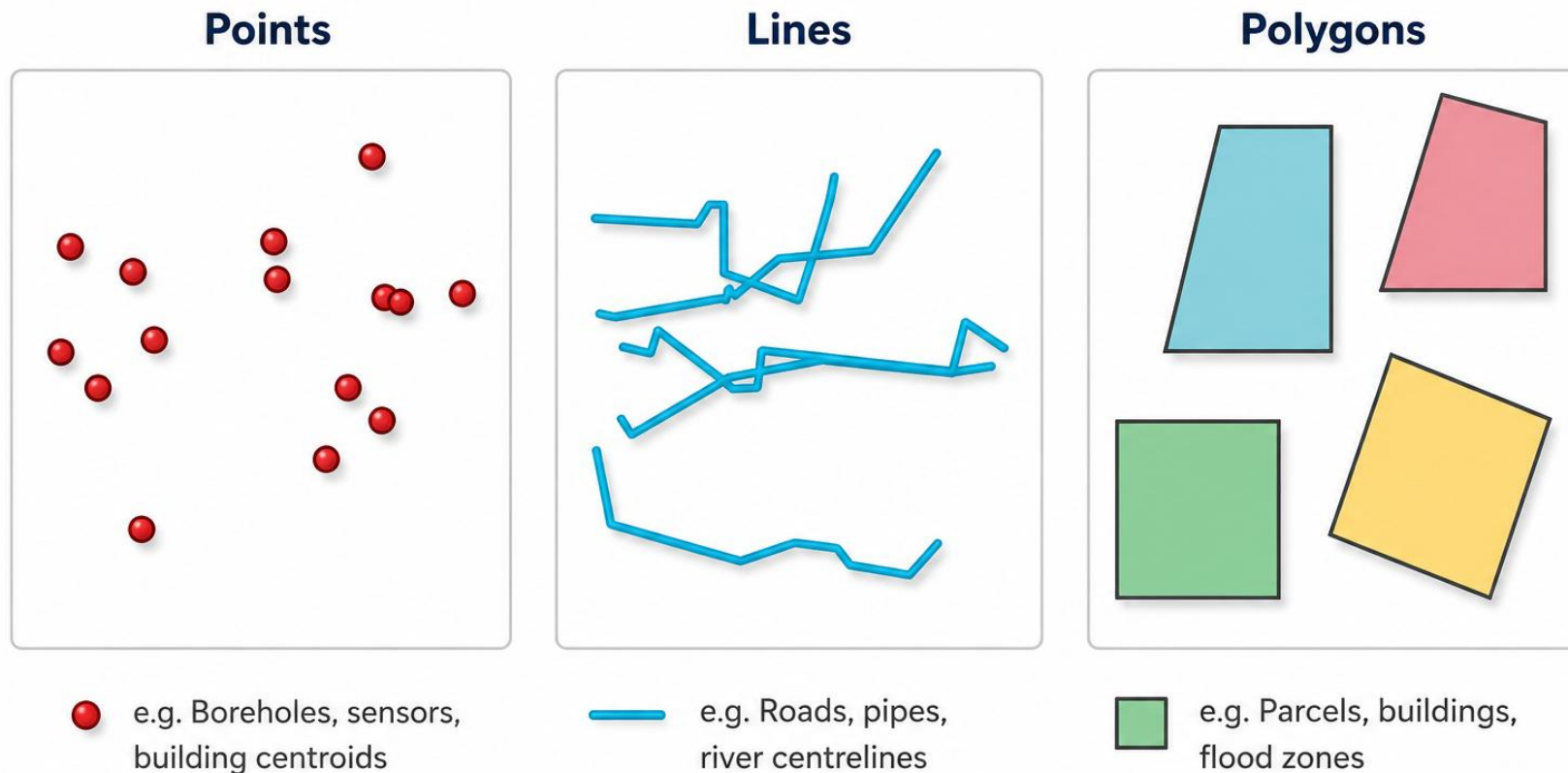
This course teaches you to execute each step using Python.

**Always ask: 'what question is this map answering?'**



*(how do we actually represent the world in a computer?)*

## Vector Data Model

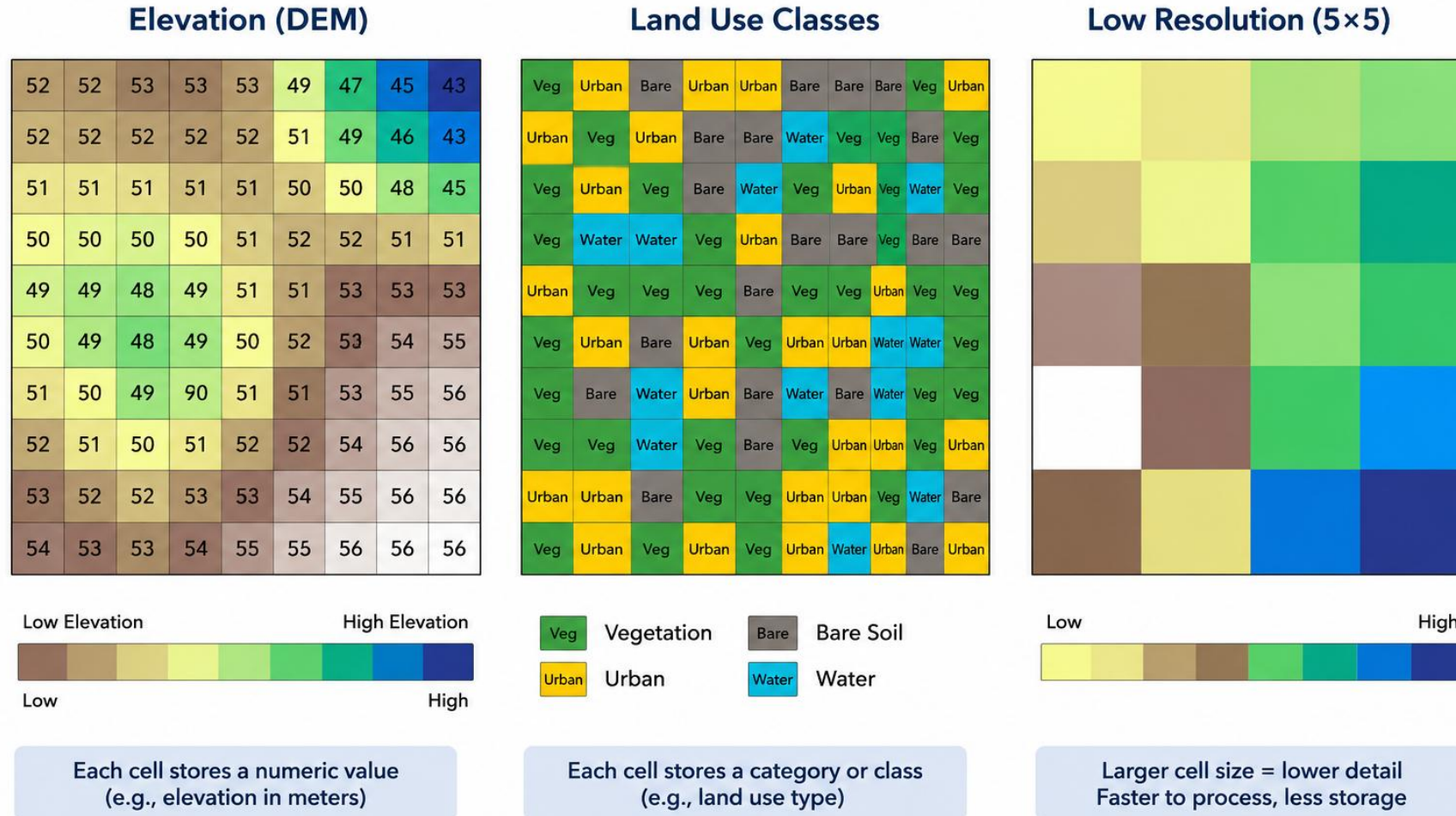


Represents discrete features with exact geometry + attributes

Each feature = a row in a table with a geometry column (GeoDataFrame)

*(how do we actually represent the world in a computer?)*

## Raster Data Model - Grids of Cells with Values



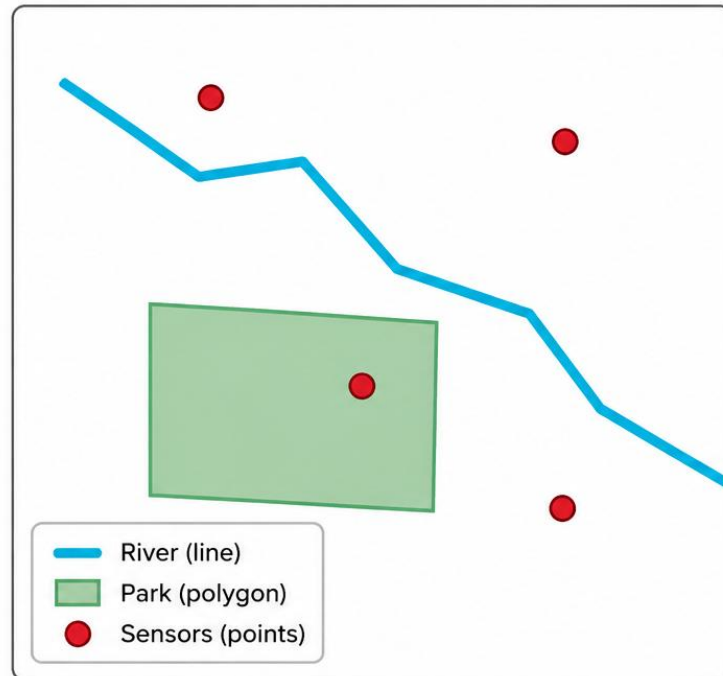
Represents continuous fields as a grid of cells (pixels), each with a value

Resolution = cell size — determines detail vs. file size trade-off

# Vector vs Raster

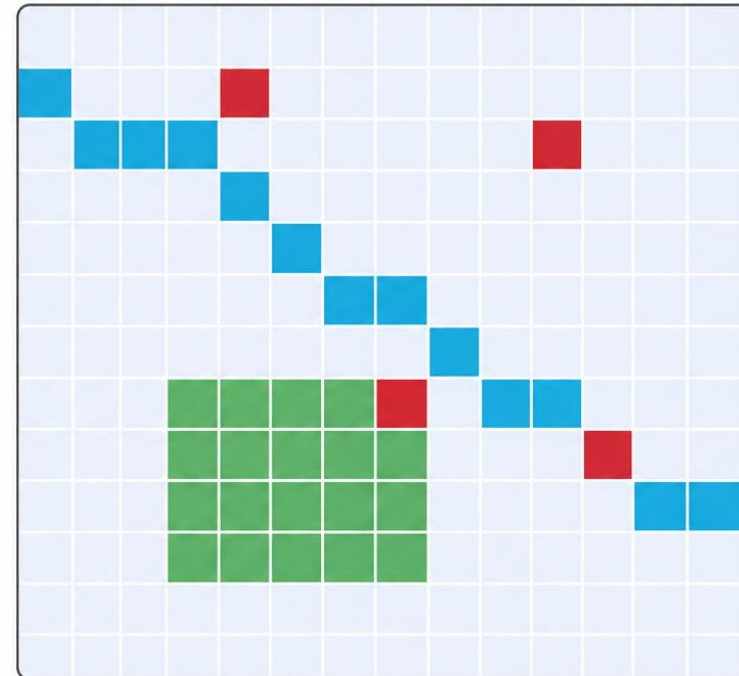
## Same Landscape — Two Different Models

### Vector Representation



*Exact geometry*  
*Small file size for sparse data*  
*Best for discrete features*

### Raster Representation



*Grid approximation*  
*Fixed cell size = detail trade-off*  
*Best for continuous fields*

**Choosing the right model depends on: what you're mapping + what analysis you need**



# LIVE DEMO

[L01\\_LayerExplorer.html](#)

Toggle spatial layers on/off to see how combining data reveals engineering insights that no single dataset can provide.

Scan to open the demo on your phone



[aardid.github.io/uc\\_205\\_gis\\_course/L01\\_LayerExplorer.html](https://aardid.github.io/uc_205_gis_course/L01_LayerExplorer.html)

# Data Sources for NZ Engineering

## National datasets:

LINZ Data Service — parcels, topo, imagery

Stats NZ — census, deprivation index

ESNZ (former GNS Science) — earthquake, landslide hazards

ESNZ (former NIWA) — climate, hydrology, flood models

OpenStreetMap — roads, buildings, amenities

## Canterbury-specific:

Canterbury Maps — regional council portal

CCC Open Data — Christchurch city assets

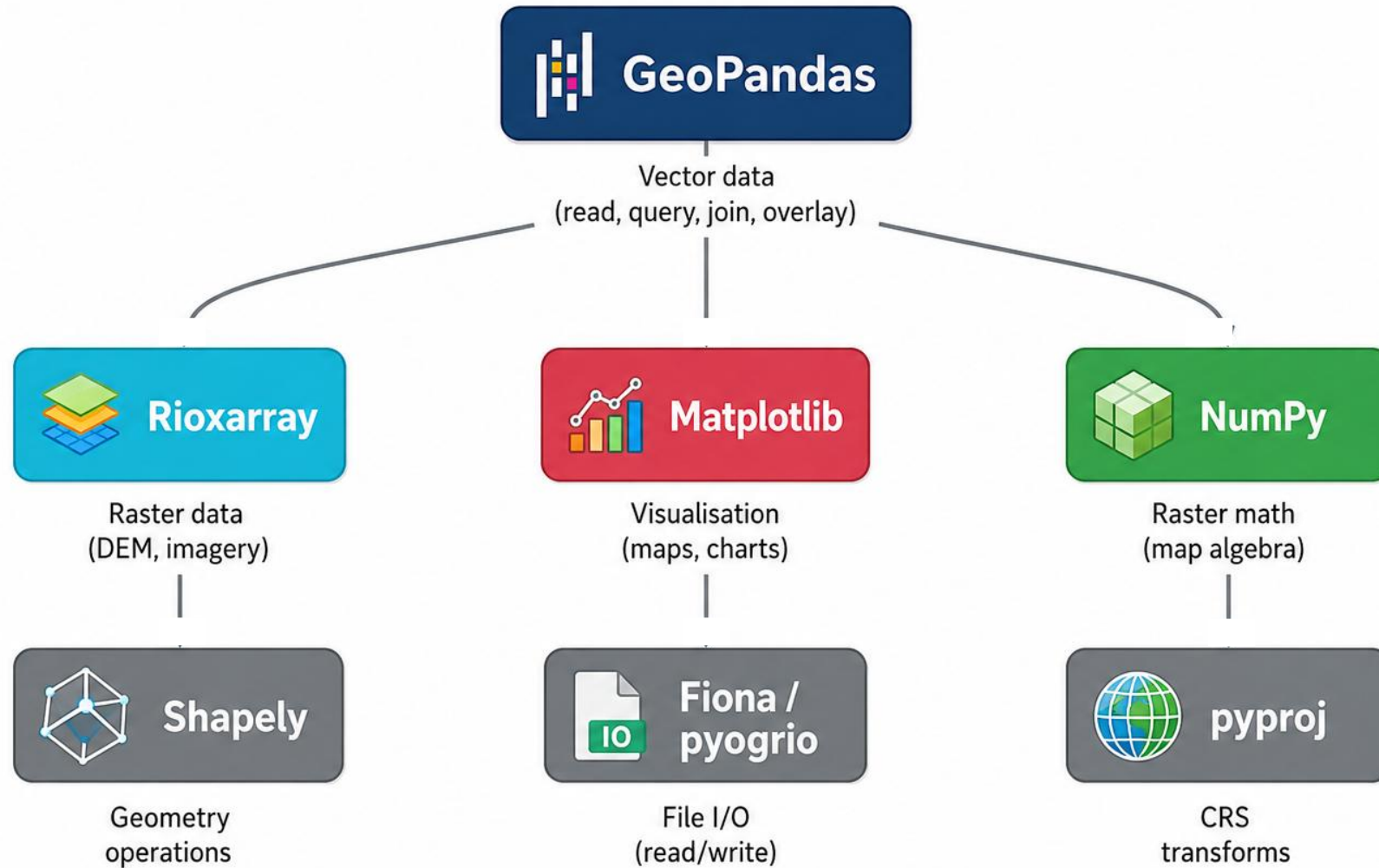
ECan — groundwater, land use, consents

Canterbury Geotechnical Database

**All freely available — no licenses needed for this course.**

**Always ask: 'who collected this - and when?'**

# The Python Geospatial Ecosystem



All open-source, all free — the same tools used by professional engineers and researchers.

# Why Python for Geospatial Engineering?

Reproducible — scripts document every step, unlike clicking through GIS menus

Scalable — process 1 file or 10,000 with the same code

Integrates with statistics — same language for spatial + statistical analysis

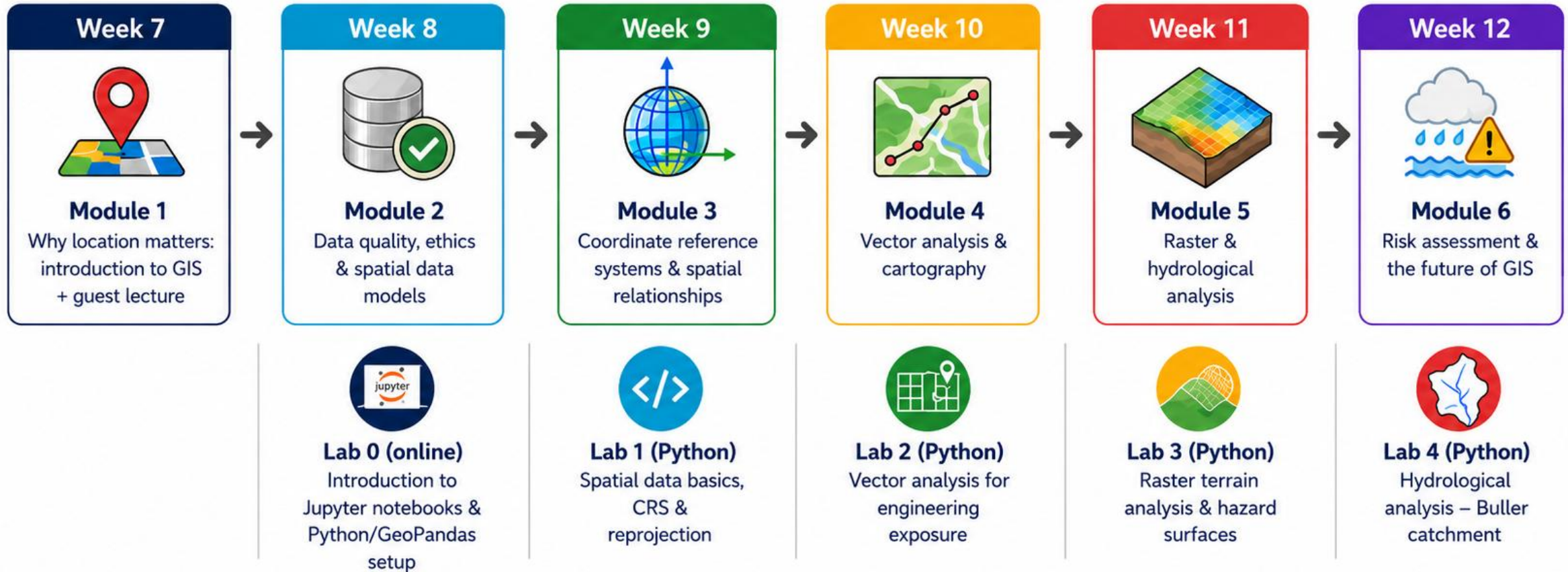
Industry standard — used at BECA, Tonkin + Taylor, AECOM, councils

Free — no license fees

**You already know Python from term 3 - now we add the spatial dimension.**



# ENCN205 GIS component – 6 Week Roadmap



| Assessment                                                                                                                 | Description                                                          | Weight (%) | Timing          | Learning Outcomes |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------|-----------------|-------------------|
|  Lab outputs & guest lecture attendance | Completion of weekly labs (Lab 0–Lab 4) and guest lecture attendance | 5          | Weeks 7–11      | 7–10              |
|  GIS Final Exam                         | Comprehensive final exam covering all GIS modules                    | 30         | To be announced | 7–10              |

# Assessment

## Weekly Sign-Off labs - 5%

8 MCQ, pass mark 8/10, unlimited retries

Covers that week's lab

Download a sign-off PNG and upload to Learn

## Written Exam - 30%

2 hours, 6 questions, pen and paper

Each question: 10 marks (60 total).

**Study tool available - 350 practice MCQs with instant feedback**

## KEY CONCEPTS

# Today's Takeaways

1. GIS is the framework that connects spatial data to engineering decisions
2. Vector = discrete features (points, lines, polygons) with attributes
3. Raster = continuous grids of cell values (elevation, rainfall, land use)
4. Choosing the right data model depends on what you're mapping
5. Python's geospatial ecosystem (GeoPandas + friends) gives you reproducible, scalable GIS

## NEXT LECTURE

### Lecture 2: Data Quality, Ethics & Professional Practice

## By the End of This Course, You Will Be Able To:

- ✓ Load, transform, and query spatial data in Python
- ✓ Choose and apply appropriate CRS for NZ engineering
- ✓ Perform vector analysis: buffers, spatial joins, overlays
- ✓ Perform raster analysis: terrain, flood modelling, map algebra
- ✓ Design clear, professional maps following cartographic principles
- ✓ Delineate catchments and model hydrological processes
- ✓ Conduct multi-criteria site evaluation for engineering decisions

# Choosing Vector or Raster

## Use Vector when:

Features have clear boundaries

You need exact geometry (lengths, areas)

Data is sparse (not every location has a value)

Examples: buildings, pipes, roads, parcels, zone boundaries

## Use Raster when:

The phenomenon varies continuously

You need cell-based maths (map algebra)

Data comes from remote sensing or models

Examples: elevation (DEM), temperature, rainfall, satellite imagery



**Think of an engineering project you've seen or worked on.  
What spatial data would you need? Vector, raster, or both?**

Examples to consider:

Designing a new subdivision — what layers do you need?

Assessing flood risk for a bridge — vector, raster, or both?

Planning a cycle path network across the city